Top 75 TMUA Mistakes

A tight, student-facing mistake bank for TMUA. The aim is not just to read this: tag every wrong answer after timed practice, find your repeat errors, and drill the exact fix.

How to use this: after every timed paper, label each wrong answer with one mistake number. If the same mistake appears in 3 sittings, stop doing random practice and drill that one error type.

Domain
Zero, negatives, logs, square roots, denominators.

Sign
Inequalities, indices, brackets, trig quadrants.

Graph
Transformations, intersections, area, sketching.

Logic
If, only if, converse, contrapositive, quantifiers.

MCQ
Trap answers, edge cases, timing, elimination.

The Big 10 - check these first

Rank	Mistake	Quick exam check
1	Dividing by a variable and deleting the zero case	Could the thing I cancelled be zero?
2	Forgetting $\sqrt{x^2} = \lvert x \rvert$	Did a square root hide a sign?
3	Not flipping an inequality after multiplying/dividing by a negative	Did I divide by something negative or possibly negative?
4	Confusing converse and contrapositive	Which arrow is actually being used?
5	Wrong negation of “if A, then B”	Negation is A true and B false.
6	Missing graph transformation reversal	$f(x+a)$ moves left, not right.
7	Confusing signed integral with area	Did the graph cross the axis?
8	Missing trig solutions in a given interval	Did I check all quadrants/periods?
9	Using examples as proof	Does this prove every case or just test some cases?
10	Forgetting edge cases: $0, 1, -1$, fractions, equality	Have I tested the annoying small cases?

1. Timed paper
75 minutes strict. No pausing.

2. Tag errors
Use mistake numbers, not vague notes.

3. Drill repeats
Same error 3 times = targeted practice.

4. Re-test
Target: fewer repeat mistakes next sitting.

1-8. Algebra and domain traps

1 Dividing by a variable without checking zero

P1 HIGH COST

TRAP: $x^2 = 5x \Rightarrow x = 5$, losing $x = 0$.

FIX: Move all terms to one side and factorise: $x^2 - 5x = x(x-5) = 0$.

2 Forgetting $\sqrt{x^2} = |x|$, not x

P1 HIGH COST

TRAP: $\sqrt{(x-3)^2} = x - 3$.

FIX: Write $\sqrt{A^2} = |A|$. Then split cases if needed.

3 Not flipping inequalities after negatives

P1 HIGH COST

TRAP: $-2x > 6 \Rightarrow x > -3$.

FIX: Dividing by a negative flips the sign: $x < -3$. If sign is variable, split cases.

4 Multiplying an inequality by an expression of unknown sign

P1

TRAP: From $\frac{1}{x} > 2$, multiplying by x without knowing sign.

FIX: Use a sign chart or bring to one side: $\frac{1-2x}{x} > 0$.

5 Splitting square roots over addition

P1

TRAP: $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.

FIX: Roots distribute over products, not sums: $\sqrt{ab} = \sqrt{a}\sqrt{b}$, but $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$.

6 Forgetting excluded values in rational expressions

P1

TRAP: Cancelling $\frac{x(x-2)}{x}$ and allowing $x = 0$.

FIX: Before cancelling, write restrictions: denominator $\neq 0$.

7 Solving after squaring without checking answers

P1

TRAP: Squaring both sides creates an extra solution.

FIX: Every time you square both sides, substitute final answers back into the original equation.

8 Ignoring log and root domains

P1 HIGH COST

TRAP: Solving $\log(x-2)$ while allowing $x \leq 2$.

FIX: Write domain first: log input > 0; even root input ≥ 0; denominator $\neq 0$.

9-16. Indices, quadratics, logs and sequences

9 Confusing index laws

P1 HIGH COST

TRAP: $a^m a^n = a^{mn}$ or $(a^m)^n = a^{m+n}$.

FIX: Same-base multiply = add powers. Power of a power = multiply powers.

10 Negative index sign confusion

P1

TRAP: $a^{-2} = -a^2$.

FIX: Negative power means reciprocal: $a^{-2} = \frac{1}{a^2}$, not a negative number.

11 Wrong discriminant or wrong conclusion from it

P1

TRAP: Using $b^2 + 4ac$, or saying $\Delta < 0$ has two roots.

FIX: $\Delta = b^2 - 4ac$. > 0: two real; $= 0$: repeated; < 0: no real roots.

12 Completing the square with the wrong sign

P1

TRAP: $x^2 + 6x + 5 = (x+3)^2 + 5$.

FIX: Add inside, compensate outside: $x^2 + 6x + 5 = (x+3)^2 - 4$.

13 Using log laws without checking positivity

P1

TRAP: $\log x + \log(x-3)$ while accepting $x = 1$.

FIX: Before log laws, each log input must be > 0.

14 Using change of base as if required

P1

TRAP: Overcomplicating a TMUA log problem with change of base.

FIX: Prefer index form and core log laws. Change of base is not normally the intended route.

15 Finite vs infinite geometric series confusion

P1 HIGH COST

TRAP: Using $S_\infty = \frac{a}{1-r}$ for a finite series.

FIX: Finite: $S_n = \frac{a(1-r^n)}{1-r}$. Infinite only if $|r| < 1$.

16 Quadratic sequence coefficient error

P1

TRAP: Second difference equals a in $an^2 + bn + c$.

FIX: For $u_n = an^2 + bn + c$, second difference is $2a$.

17-24. Functions, graphs and coordinate geometry

17 Graph transformation reversal

P1 HIGH COST

TRAP: Thinking $f(x+3)$ shifts right by 3.

FIX: Inside is opposite: $f(x+3)$ moves left 3. $f(ax)$ has horizontal factor $\frac{1}{a}$.

18 Confusing $af(x)$ with $f(ax)$

P1

TRAP: Treating vertical stretch and horizontal stretch as the same.

FIX: Outside affects y-values. Inside affects x-values in the opposite way.

19 Forgetting function order in composition

P1

TRAP: $f(g(x))$ means do f first.

FIX: $f(g(x))$: do g first, then feed result into f.

20 Intersections: solving only one graph

P1

TRAP: Finding x-intercepts when the question asks where two graphs meet.

FIX: For intersections, set equations equal to each other, not equal to zero unless one graph is the axis.

21 Expanded circle centre sign error

P1 HIGH COST

TRAP: $x^2+y^2+2gx+2fy+c=0$ has centre (g, f).

FIX: Centre is $(-g, -f)$. Complete the square to confirm.

22 Tangent/radius gradient confusion

P1

TRAP: Tangent gradient equals radius gradient.

FIX: Tangent is perpendicular to radius, so $m_{\text{tan}}m_{\text{rad}} = -1$.

23 Line gradient sign error

P1

TRAP: Using $\frac{x_2-x_1}{y_2-y_1}$ for gradient.

FIX: Gradient is change in y over change in x: $m = \frac{y_2-y_1}{x_2-x_1}$.

24 Approximate graph solution overread

P1

TRAP: Reading exact answers from a sketch.

FIX: Use sketch for structure and intervals; use algebra for exact values.

25-34. Geometry, trigonometry, radians and vectors

25 Using the wrong circle theorem

P1

TRAP: Confusing same segment, cyclic quadrilateral, and alternate segment.

FIX: Mark chord/tangent/centre first, then choose theorem.

26 Angle at centre ratio reversed

P1

TRAP: Circumference angle is twice centre angle.

FIX: Centre angle is twice circumference angle on the same arc.

27 Similarity scale factor error

P1

TRAP: Using k for length, area and volume.

FIX: Length k, area k^2, volume k^3.

28 Sine rule ambiguous case missed

P1 HIGH COST

TRAP: Assuming SSA gives only one possible triangle.

FIX: If using sine rule to find an angle, check θ and $180^\circ - \theta$.

29 Missing trig solutions in an interval

P1 HIGH COST

TRAP: Finding the principal solution only.

FIX: Use symmetry/periodicity and test all solutions in the required interval.

30 Exact trig value mix-up

P1

TRAP: Swapping $\sin 30^\circ$ and $\cos 30^\circ$.

FIX: Use the 30-60-90 and 45-45-90 triangles.

31 Radian/degree confusion

P1 HIGH COST

TRAP: Using $s = r\theta$ when θ is in degrees.

FIX: Arc and sector formulae need radians: $s = r\theta$, $A = \frac{1}{2}r^2\theta$.

32 Sector and segment mix-up

P1

TRAP: Giving sector area when the question asks for segment area.

FIX: Segment = sector - triangle.

33 Vector parallel condition incomplete

P1

TRAP: Comparing only one component.

FIX: Parallel vectors need a common scalar multiple: $\mathbf{a} = k\mathbf{b}$.

34 Midpoint/vector average error

P1

TRAP: Adding coordinates without dividing by 2.

FIX: $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$, or $\mathbf{m} = \frac{\mathbf{a}+\mathbf{b}}{2}$.

35-44. Calculus, area, number and probability/statistics

35 Differentiating before simplifying

P1

TRAP: Applying calculus to a messy fraction instead of simplifying first.

FIX: Expand, cancel, divide through, and rewrite as powers before differentiating.

36 Forgetting $+C$ in indefinite integration

P1

TRAP: Losing the constant needed for $y(1) = 3$.

FIX: Write $+C$ immediately. Use the given point to find it.

37 Area vs signed integral

P1 HIGH COST

TRAP: Treating $\int_a^b f(x)dx$ as area when graph goes below axis.

FIX: Split at roots and add absolute values of each region.

38 Trapezium rule endpoint error

P1

TRAP: Doubling all ordinates.

FIX: First and last ordinates are single; middle ordinates are doubled.

39 Direct/inverse proportion setup error

P1

TRAP: Writing $y = kx$ when $y \propto \frac{1}{x}$.

FIX: Direct: $y = kx^n$. Inverse: $y = \frac{k}{x^n}$.

40 Bounds and rounding treated as exact

P1

TRAP: Using rounded measurements as exact values.

FIX: Nearest unit means true value lies in $[x - 0.5, x + 0.5)$.

41 Independence vs mutually exclusive

P1

TRAP: Thinking mutually exclusive means independent.

FIX: Independent: $P(A \cap B) = P(A)P(B)$. Mutually exclusive: $P(A \cap B) = 0$.

42 Conditional probability denominator error

P1 HIGH COST

TRAP: Dividing by total sample space instead of the given condition.

FIX: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Denominator is always the “given”.

43 Tree diagram without-replacement error

P1

TRAP: Keeping second-branch probabilities unchanged after an item is removed.

FIX: If no replacement, update both numerator and denominator.

44 Histogram frequency-density error

P1

TRAP: Bar height = frequency when class widths differ.

FIX: Bar height is frequency density: $\frac{\text{frequency}}{\text{class width}}$.

45-56. Logic, implication and language traps

45 Confusing converse with contrapositive

P2 HIGH COST

TRAP: Treating $Q \Rightarrow P$ as equivalent to $P \Rightarrow Q$.

FIX: Only contrapositive is equivalent: $P \Rightarrow Q$ equals $\neg Q \Rightarrow \neg P$.

46 Necessary and sufficient backwards

P2 HIGH COST

TRAP: If $P \Rightarrow Q$, saying P is necessary.

FIX: P is sufficient for Q. Q is necessary for P.

47 "A if B" vs "A only if B" reversal

P2

TRAP: Reading both as the same arrow.

FIX: "A if B" means $B \Rightarrow A$. "A only if B" means $A \Rightarrow B$.

48 Wrong negation of "if A then B"

P2 HIGH COST

TRAP: Negating $A \Rightarrow B$ as $\neg A \Rightarrow \neg B$.

FIX: Negation is: A is true and B is false.

49 Wrong quantifier negation

P2

TRAP: "Not all" means "none".

FIX: Not all = at least one not. Not exists = none.

50 Treating "or" as exclusive

P2

TRAP: Thinking "A or B" excludes both.

FIX: In TMUA logic, "or" is inclusive unless stated otherwise.

51 Confusing "if" with "if and only if"

P2

TRAP: Proving one direction and claiming equivalence.

FIX: Iff needs both directions: $A \Rightarrow B$ and $B \Rightarrow A$.

52 Affirming the consequent

P2 HIGH COST

TRAP: $P \Rightarrow Q$, Q true, therefore P true.

FIX: This is invalid. Q may happen for other reasons.

53 Denying the antecedent

P2

TRAP: $P \Rightarrow Q$, P false, therefore Q false.

FIX: Invalid. The implication says nothing when P is false.

54 Misreading "must be true" vs "could be true"

P2 HIGH COST

TRAP: Selecting an option that is possible but not forced.

FIX: Underline must/could/may/always/never before doing maths.

55 Assuming the domain

P2

TRAP: Treating integers, positive integers, reals, and non-zero reals as the same.

FIX: Write the domain before testing examples or counterexamples.

56 "Not enough information" confused with false

P2

TRAP: If a conclusion does not follow, calling it false.

FIX: Separate "not implied" from "false". A claim can be true but not proved by the premises.

57-66. Proof, counterexample and reasoning traps

57 Thinking multiple counterexamples are needed

P2

TRAP: Trying to disprove "for all" with many examples.

FIX: One valid counterexample disproves a universal statement.

58 Using examples to prove a universal claim

P2 HIGH COST

TRAP: Testing $n = 1, 2, 3$ and saying "therefore true".

FIX: Examples suggest a conjecture; they do not prove all cases.

59 Missing edge cases

P2 HIGH COST

TRAP: Ignoring $0, 1, -1$, equality cases, fractions, and boundary values.

FIX: Before algebra, test the "annoying five": $0, 1, -1$, fractions, boundary.

60 Division by zero inside proof analysis

P2

TRAP: A false proof hides $a - b = 0$ in a cancellation step.

FIX: Every cancellation in a proof must have a non-zero justification.

61 Assuming $f(A) = f(B) \Rightarrow A = B$

P2

TRAP: From $x^2 = 4$, concluding $x = 2$.

FIX: This only works for one-to-one functions. Many functions are not injective.

62 Proving the wrong direction

P2

TRAP: Asked $P \Rightarrow Q$, proving $Q \Rightarrow P$.

FIX: Write start and finish: "Assume P. Need Q."

63 Incomplete proof by cases

P2

TRAP: Proving the even case but forgetting the odd case.

FIX: Cases must be exhaustive and non-overlooked.

64 Proof-order sequencing error

P2 HIGH COST

TRAP: Choosing a true statement that does not follow next.

FIX: Each proof line must be justified by earlier lines, not just be true in isolation.

65 Contradiction proof starts with the wrong assumption

P2

TRAP: To prove S, assuming something unrelated instead of $\neg S$.

FIX: Contradiction begins with "Assume not S" and ends by rejecting that assumption.

66 Confusing conjecture with proof

P2

TRAP: Spotting a pattern and stopping there.

FIX: A conjecture is a guess. You still need a general argument or counterexample.

67-75. MCQ, timing and exam-process mistakes

67 Over-solving when elimination is faster

P1 P2

TRAP: Doing a full derivation when two options can be killed by sign/size.

FIX: In MCQ, first ask: can I eliminate by domain, sign, units, or edge case?

68 Not using no-negative-marking properly

P1 P2

TRAP: Leaving blanks or refusing to guess.

FIX: Always select an answer. Eliminate, guess, flag, move.

69 Spending 8 minutes on one question

P1 P2 HIGH COST

TRAP: Losing easy later marks because one hard question became personal.

FIX: At 2.5-3 minutes with no route: guess, flag, move.

70 Answering the question you expected, not the one asked

P1 P2

TRAP: Finding x when the question asks for $2x + 1$.

FIX: Before selecting, reread the final sentence and units.

71 Not checking answer-choice scale

P1

TRAP: Choosing 1000 when a rough estimate is 10.

FIX: Do a 5-second magnitude estimate before committing.

72 Units and compound-unit conversion error

P1

TRAP: Mixing minutes and hours, cm^2 and m^2, or density units.

FIX: Convert units before substituting into formulae.

73 Copying the wrong value from working to option

P1 P2

TRAP: Correct working, wrong selected option.

FIX: Circle the final requested quantity, not just the intermediate value.

74 Forgetting to test a trap option

P1 P2

TRAP: Your wrong working appears as an answer choice, so it feels validated.

FIX: If an answer appears too neatly, test it in the original condition.

75 Doing practice without tagging mistakes

P1 P2 HIGH COST

TRAP: More papers, same errors, no score movement.

FIX: After each paper, tag each wrong answer by mistake number and drill repeats.

Weekly TMUA error tracker

Print this page or copy it into a spreadsheet. The goal is not to collect mistakes; the goal is to stop repeating the same mistake number.

Week	Paper / test	Raw score	Top 3 mistake numbers	What I will drill	Fixed next week?
1					
2					
3					
4					
5					
6					

Rapid checklist by category

A Algebra and domain

- Move to one side before cancelling variables.
- Write denominator/log/root restrictions first.
- Check answers after squaring.
- For inequalities, ask “negative or possibly negative?”

B Graphs and geometry

- Inside graph transformations reverse direction.
- Circle centre signs reverse in expanded form.
- Radius and tangent are perpendicular.
- Radians are needed for $s = r\theta$ and sector area.

C Calculus and data

- Simplify before differentiating/integrating.
- Area is not always the signed integral.
- Conditional probability denominator is the given event.
- Histogram height is frequency density, not frequency.

D Logic and proof

- Contrapositive is equivalent; converse is not.
- “Only if” points to the necessary condition.
- One counterexample disproves “for all”.
- Examples do not prove universal statements.

Final 20-second exam protocol

Read
Underline must/could/all/some/only if.

Restrict
Domain, sign, denominator, interval.

Sketch/test
Try $0, 1, -1$, boundary, rough graph.

Select
Eliminate, choose, flag, move.

Rule: if a wrong answer keeps appearing in your working, it is not a “careless mistake”. It is a trained reflex. Replace it with a new reflex using the fix.